HW3

Wednesday, 1 March 2023 1:38 PM

Question 1

Ab-initio phase diagram for H-adsorption on Pt(111) (surface & sub-surface layers)

$$E_{ads} = \left(\underbrace{E^{nH}_{DFT} - E^{Pt(111)}_{DFT}}_{\text{XS surface energy}} - \frac{n}{2} E^{H_2}_{DFT}\right)\frac{1}{n}$$

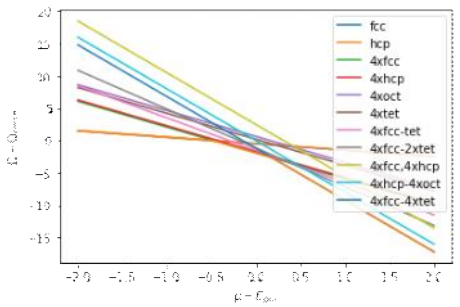
+ XS surface energy : $(E_{DFT} - E^{clean}_{DFT})$ → instead of $(\mu_{Pt} \cdot N_{Pt})$
+ XS surface entropy = 0
  - ↳ One way to better approximate this would be to calculate configurational entropy. i.e. counting all possible arrangements of H adsorbate.
+ Plot : Ω vs. μ (x-axis : $(\mu - E_{gas})$)
+ Plot : T, P scale

+ ZPE's on surface → assume each H-atom has harmonic vibrational freq. of 450, 450 and 900 cm^{-1}
+ Assuming surface is in equilibrium w/ bulk phases $[\mu_i = \mu_i^{gas}\ ,\ \mu_i = \mu_i^{bulk\,metal}]$

$$\gamma = (U - \cancel{TS} - \Sigma \mu_i N_i)\frac{1}{A} = (U - \mu_H n_H - \mu_{Pt} N_{Pt})\frac{1}{A}$$

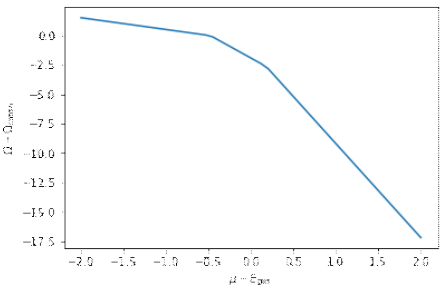
$$U \approx (n_H \cdot E_{ads}) + ZPE \quad ; \quad ZPE = \Sigma \frac{1}{2} h\nu_i$$

$$\gamma A = (E_{ads} + ZPE - \mu_H n_H) = \Omega \leftarrow \text{varying } \mu_H \text{ will give values of } \Omega$$



This phase diagram shows that certain phase plane and subsurface structures are thermodynamically stable and adsorption on these surfaces will be prefered.

↳ The lowest $(\Omega - \Omega_{clean})$ values can be extrapolated and plotted as shown on the figue on the right



T, P relations : $\mu_H = U^H_{tot} + k_B T - TS^H_{tot}$ (i.g.)

$$U^H_{tot} = U_{vib} + U_{rot} + U_{trans}$$

$$= \sum \left(\frac{1}{2} h\nu_i + \frac{h\nu_i}{e^{\beta h\nu_i} - 1}\right) + k_B T + \frac{3}{2} k_B T \quad ; \quad \beta = \frac{1}{k_B T}$$

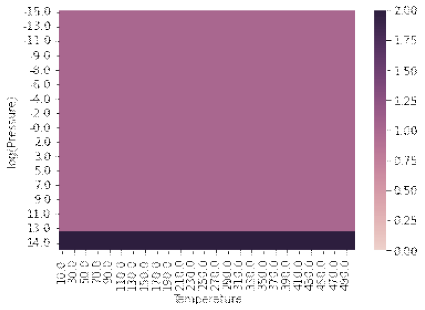
ZPE | Thermal excitation | Linear

$$S^H_{tot} = S_{vib} + S_{rot} + S_{trans}$$

$$= \sum_i -\frac{h\nu_i}{2T} - k_B \ln(1 - e^{-h\nu_i \beta}) + \frac{1}{T}\sum_i \left(\frac{1}{2} h\nu_i + \frac{h\nu_i}{e^{\beta h\nu_i} - 1}\right) + \left[k_B \ln\left[\left(\frac{m}{2\pi\beta\hbar^2}\right)^{3/2} \frac{k_B T}{P}\right] + \frac{5}{2} k_B T\right] n_H + k_B \ln\left(\frac{I}{\sigma \beta \hbar^2}\right) + k_B$$

Given a T & P array, μ_H can be calculated and this value can be associated w/ the lowest $(\Omega - \Omega_{clean})$ value and then corresponding H coverage (ML).

↳ This can be used to create a heat map.



There seems to be a bug w/ my code which I could not figure out.

↓

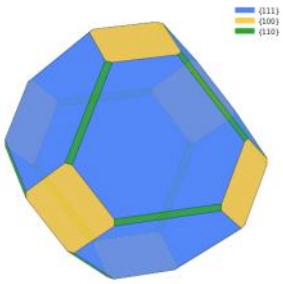
The H-ML should increase w/ pressure and this trend is shown in the graph above, however temp. variation is not explored properly

↓

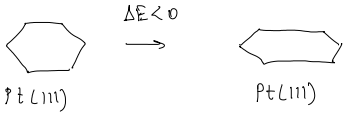
From intuition, the ML should decrease to 0 at higher temp since the system has a much higher energy

Question 2

(a) Wulff diagram is shown



(b) For a favorable interaction, the nanoparticle shape will be "wetted" i.e. become squished



(c) Shape of the nanoparticle change w/ T and P

↳ Using given T & P, a similar approach can be followed where $\Omega - \Omega_{clean}$ can be calculated to given an associated γ value which can be added to the $E_{surf.}$ to obtain new relationship.

↳ Sample calculation is shown in the attached jupyter notebook → could not figure out bug in Q1 which inhibits me from answering this question